第 13 章　多元函数微分学

1.【答案】 $\frac{\pi}{2}\mathrm{d}x$

【解析】 $\frac{\partial z}{\partial x}=\frac{y-\sin(x+y)}{1+\left[xy+\cos(x+y)\right]^2}$，$\frac{\partial z}{\partial y}=\frac{x-\sin(x+y)}{1+\left[xy+\cos(x+y)\right]^2}$，

故 $$\mathrm{d}z\big|_{(0,\pi)}=\frac{\partial z}{\partial x}\Big|_{(0,\pi)}\mathrm{d}x+\frac{\partial z}{\partial y}\Big|_{(0,\pi)}\mathrm{d}y=\frac{\pi}{2}\mathrm{d}x\,.$$

2.【答案】 $\frac{y}{\sin x}+\frac{x}{\sin y}$

【解析】 因为 $z=f(\cos y-\cos x)+xy$，所以

$$\frac{\partial z}{\partial x}=f'(\cos y-\cos x)\cdot\sin x+y\,,\ \frac{\partial z}{\partial y}=f'(\cos y-\cos x)\cdot(-\sin y)+x\,.$$

故 $$\frac{1}{\sin x}\cdot\frac{\partial z}{\partial x}+\frac{1}{\sin y}\cdot\frac{\partial z}{\partial y}=\frac{y}{\sin x}+\frac{x}{\sin y}\,.$$

3.【答案】 $2y^3x^{y^2}(1+\ln x)f'(x^{y^2})+yf(x^{y^2})$

【解析】 $\frac{\partial z}{\partial x}=yf'(x^{y^2})\cdot(y^2x^{y^2-1})$，$\frac{\partial z}{\partial y}=f(x^{y^2})+yf'(x^{y^2})\cdot(x^{y^2}\cdot\ln x\cdot 2y)$，

所以 $$2x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=2y^3x^{y^2}(1+\ln x)f'(x^{y^2})+yf(x^{y^2})\,.$$

4.【答案】 $\frac{1}{5}$

【解析】 将 $x=0$，$y=2$ 代入 $(x+1)z+2y\ln z-\arctan(xy)=1$，得 $z+4\ln z=1$．因为 $z+4\ln z$ 是 $(0,+\infty)$ 上的增函数，且当 $z=1$ 时，$z+4\ln z=1$，从而解得 $z=1$．

$(x+1)z+2y\ln z-\arctan(xy)=1$ 两端分别对 x 求偏导，得

$$z+(x+1)\frac{\partial z}{\partial x}+2y\cdot\frac{1}{z}\cdot\frac{\partial z}{\partial x}-\frac{1}{1+(xy)^2}\cdot y=0\,,$$

将 $x=0$，$y=2$，$z=1$ 代入上式，解得 $\frac{\partial z}{\partial x}\Big|_{(0,2)}=\frac{1}{5}$．

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5.【答案】 $2e^{x-y}$

【解析】 $$F(x,y)=\int_0^{x-y}(x-y-t)e^t dt=(x-y)\int_0^{x-y}e^t dt-\int_0^{x-y}te^t dt,$$

于是 $$\frac{\partial F}{\partial x}=\int_0^{x-y}e^t dt+(x-y)e^{x-y}-(x-y)e^{x-y}=\int_0^{x-y}e^t dt,\ \frac{\partial^2 F}{\partial x^2}=e^{x-y},$$

$$\frac{\partial F}{\partial y}=-\int_0^{x-y}e^t dt-(x-y)e^{x-y}+(x-y)e^{x-y}=-\int_0^{x-y}e^t dt,\ \frac{\partial^2 F}{\partial y^2}=e^{x-y},$$

故 $$\frac{\partial^2 F}{\partial x^2}+\frac{\partial^2 F}{\partial y^2}=e^{x-y}+e^{x-y}=2e^{x-y}.$$

6.【答案】 -1

【解析】 由 $f'_x(x,y)=1+2\cos x$，知 $f(x,y)=x+2\sin x+g(y)$.

由 $f(x,\sin x)=x+\sin x$，得 $g(\sin x)=-\sin x$，即 $g(y)=-y$，故

$$f(x,y)=x+2\sin x-y,$$

则 $f'_y(x,y)\big|_{y=\sin x}=-1\big|_{y=\sin x}=-1$.

7.【答案】 $\frac{\pi^2}{4}+2$

【解析】 由 $\frac{\partial^2[f(x,y)]}{\partial x\partial y}=1$，得 $\frac{\partial[f(x,y)]}{\partial y}=x+C_1(y)$，所以

$$f(x,y)=xy+\int C_1(y)dy+C_2(x)=xy+C_3(y)+C_2(x).$$

由 $f(0,y)=\sin y$，$f(x,0)=\sin x$，得 $C_2(x)=\sin x-C_3(0)$，$C_3(y)=\sin y-C_2(0)$.

又 $f(0,0)=0$，所以 $f(x,y)=xy+\sin x+\sin y$. 因此 $f\left(\frac{\pi}{2},\frac{\pi}{2}\right)=\frac{\pi^2}{4}+2$.

8.【答案】 $\frac{2x(y^2-y+1)}{(1+y)^2}$

【解析】 令 $u=x+y$，$v=\frac{x}{y}$，则 $x=\frac{uv}{1+v}$，$y=\frac{u}{1+v}$，于是

$$f(u,v)=\left(\frac{uv}{1+v}\right)^2-\frac{u^2v}{(1+v)^2}+\frac{u^2}{(1+v)^2},$$

即 $$f(x,y)=\frac{x^2}{(1+y)^2}(y^2-y+1),$$

所以 $f'_x(x,y)=\frac{2x(y^2-y+1)}{(1+y)^2}$.

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9.【解析】 $f(x,y)=\begin{cases}\sqrt{x|y|}, & (x,y)\in D_1=\{(x,y)\,|\,x>0,-\infty<y<+\infty\},\\ \sqrt{-x|y|}, & (x,y)\in D_2=\{(x,y)\,|\,x<0,-\infty<y<+\infty\},\\ 0, & (x,y)\in D_3=\{(x,y)\,|\,x=0,-\infty<y<+\infty\}.\end{cases}$

①当 $x=0$，$y=0$ 时，$f'_x(0,0)=\lim\limits_{\Delta x\to 0}\dfrac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim\limits_{\Delta x\to 0}\dfrac{\sqrt{|\Delta x|\cdot 0}-0}{\Delta x}=0$；

②当 $x=0$，$y\neq 0$ 时，$f'_x(0,y)=\lim\limits_{\Delta x\to 0}\dfrac{f(\Delta x,y)-f(0,y)}{\Delta x}=\lim\limits_{\Delta x\to 0}\dfrac{\sqrt{|\Delta x|}\cdot\sqrt{|y|}}{\Delta x}$，该极限不存在，所以 $f'_x(0,y)(y\neq 0)$ 不存在；

③当 $(x,y)\in D_1$ 时，$f'_x(x,y)=\dfrac{\partial}{\partial x}\left(\sqrt{x|y|}\right)=\dfrac{\sqrt{|y|}}{2\sqrt{x}}=\dfrac{1}{2}\sqrt{\left|\dfrac{y}{x}\right|}$；

④当 $(x,y)\in D_2$ 时，$f'_x(x,y)=\dfrac{\partial}{\partial x}\left(\sqrt{-x|y|}\right)=-\dfrac{\sqrt{|y|}}{2\sqrt{-x}}=-\dfrac{1}{2}\sqrt{\left|\dfrac{y}{x}\right|}$.

综上，$f'_x(x,y)=\begin{cases}\dfrac{1}{2}\sqrt{\left|\dfrac{y}{x}\right|}, & x>0,-\infty<y<+\infty,\\ -\dfrac{1}{2}\sqrt{\left|\dfrac{y}{x}\right|}, & x<0,-\infty<y<+\infty,\\ 0, & x=0,y=0,\\ \text{不存在}, & x=0,y\neq 0.\end{cases}$

10.【答案】 (B)

【解析】 设该二元函数为 $z=z(x,y)$，由题意知，$\dfrac{\partial z}{\partial y}=\dfrac{x}{y^2}$，可得 $z(x,y)=-\dfrac{x}{y}+\varphi(x)$，此时有

$$\frac{\partial z}{\partial x}=-\frac{1}{y}+\varphi'(x)=P(x,y),$$

观察选项可知，只有选项(B)符合题意.

【注】 (D)项不能选，因为 $\dfrac{1}{x^2}-\dfrac{1}{y}$ 在 $(0,y)$ 处无定义.

11.【答案】 0

【解析】 由题设 $\dfrac{\partial u}{\partial x}=\varphi'(u)\dfrac{\partial u}{\partial x}+P(x)$，$\dfrac{\partial u}{\partial y}=\varphi'(u)\dfrac{\partial u}{\partial y}-P(y)$，所以

$$\frac{\partial u}{\partial x}=\frac{P(x)}{1-\varphi'(u)},\quad \frac{\partial u}{\partial y}=-\frac{P(y)}{1-\varphi'(u)}.$$

因此

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$$P(x)\frac{\partial z}{\partial y}+P(y)\frac{\partial z}{\partial x}=P(x)f'(u)\frac{\partial u}{\partial y}+P(y)f'(u)\frac{\partial u}{\partial x}$$

$$=-\frac{P(x)P(y)f'(u)}{1-\varphi'(u)}+\frac{P(x)P(y)f'(u)}{1-\varphi'(u)}=0.$$

12. 【答案】$f_2'+2xf_{11}''+(2x^2+y)f_{12}''+xyf_{22}''$

【解析】由题可得 $u_x'=2xf_1'+yf_2'$，

因为f具有二阶连续偏导数，所以$f_{21}''=f_{12}''$，故

$$u_{xy}''=2xf_{11}''+2x^2f_{12}''+f_2'+yf_{21}''+xyf_{22}''=f_2'+2xf_{11}''+(2x^2+y)f_{12}''+xyf_{22}''.$$

13. 【解析】函数复合关系如图所示，所以

$$\frac{\partial g}{\partial x}=y-f_u'\cdot\left(-\frac{y}{x^2}\right)-f_v'\cdot\frac{1}{y},\ \frac{\partial g}{\partial y}=x-f_u'\cdot\frac{1}{x}-f_v'\cdot\left(-\frac{x}{y^2}\right),$$

$$\frac{\partial^2 g}{\partial x^2}=\frac{y}{x^2}\left[f_{uu}''\cdot\left(-\frac{y}{x^2}\right)+f_{uv}''\cdot\frac{1}{y}\right]+f_u'\cdot\left(-\frac{2y}{x^3}\right)-\frac{1}{y}\left[f_{vu}''\cdot\left(-\frac{y}{x^2}\right)+f_{vv}''\cdot\frac{1}{y}\right]$$

$$=-\frac{y^2}{x^4}f_{uu}''+\frac{2}{x^2}f_{uv}''-\frac{1}{y^2}f_{vv}''-\frac{2y}{x^3}f_u',$$

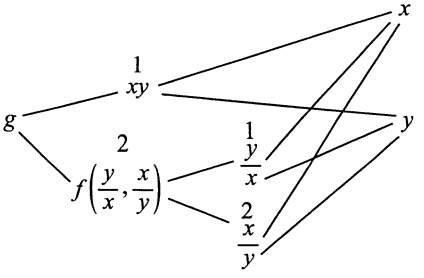
$$\frac{\partial^2 g}{\partial x\partial y}=1+f_u'\cdot\frac{1}{x^2}+\frac{y}{x^2}\left[f_{uu}''\cdot\frac{1}{x}+f_{uv}''\cdot\left(-\frac{x}{y^2}\right)\right]+\frac{1}{y^2}f_v'-\frac{1}{y}\left[f_{vu}''\cdot\frac{1}{x}+f_{vv}''\cdot\left(-\frac{x}{y^2}\right)\right]$$

$$=1+\frac{y}{x^3}f_{uu}''-\frac{2}{xy}f_{uv}''+\frac{x}{y^3}f_{vv}''+\frac{1}{x^2}f_u'+\frac{1}{y^2}f_v',$$

$$\frac{\partial^2 g}{\partial y^2}=-\frac{1}{x}\left[f_{uu}''\cdot\frac{1}{x}+f_{uv}''\cdot\left(-\frac{x}{y^2}\right)\right]-\frac{2x}{y^3}f_v'+\frac{x}{y^2}\left[f_{vu}''\cdot\frac{1}{x}+f_{vv}''\cdot\left(\frac{x}{-y^2}\right)\right]$$

$$=-\frac{1}{x^2}f_{uu}''+\frac{2}{y^2}f_{uv}''-\frac{x^2}{y^4}f_{vv}''-\frac{2x}{y^3}f_v',$$

故 $$x^2\frac{\partial^2 g}{\partial x^2}+2xy\frac{\partial^2 g}{\partial x\partial y}+y^2\frac{\partial^2 g}{\partial y^2}=2xy.$$



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14.【解析】(1)因为 $z=xyf\left(\frac{y}{x}\right)$，且 $f(u)$ 可导，所以

$$\frac{\partial z}{\partial x}=yf\left(\frac{y}{x}\right)+xyf'\left(\frac{y}{x}\right)\cdot\left(-\frac{y}{x^2}\right)=yf\left(\frac{y}{x}\right)-\frac{y^2}{x}f'\left(\frac{y}{x}\right),$$

$$\frac{\partial z}{\partial y}=xf\left(\frac{y}{x}\right)+xyf'\left(\frac{y}{x}\right)\cdot\frac{1}{x}=xf\left(\frac{y}{x}\right)+yf'\left(\frac{y}{x}\right),$$

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=xyf\left(\frac{y}{x}\right)-y^2f'\left(\frac{y}{x}\right)+xyf\left(\frac{y}{x}\right)+y^2f'\left(\frac{y}{x}\right)=2xyf\left(\frac{y}{x}\right).$$

又 $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=y^2(\ln y-\ln x)$，所以 $2xyf\left(\frac{y}{x}\right)=y^2(\ln y-\ln x)$，即

$$f\left(\frac{y}{x}\right)=\frac{y}{2x}\ln\frac{y}{x},$$

令 $u=\frac{y}{x}$，则 $f(u)=\frac{u}{2}\ln u$，即 $f(x)=\frac{x}{2}\ln x$.

(2)设 $f(x)$ 与 x 轴所围图形面积及该图形绕 x 轴旋转一周所得旋转体体积分别为 S，V，则

$$S=\left|\int_0^1 f(x)\mathrm{d}x\right|=\left|\int_0^1\frac{x}{2}\ln x\mathrm{d}x\right|=\left|\frac{1}{2}\int_0^1\ln x\mathrm{d}\left(\frac{1}{2}x^2\right)\right|=\left|\frac{1}{4}x^2\ln x\Big|_0^1-\frac{1}{4}\int_0^1 x\mathrm{d}x\right|=\frac{1}{8},$$

$$V=\int_0^1\pi[f(x)]^2\mathrm{d}x=\int_0^1\pi\left(\frac{x}{2}\ln x\right)^2\mathrm{d}x=\frac{\pi}{4}\int_0^1 x^2\ln^2 x\mathrm{d}x$$

$$\xlongequal{令\ln x=t}\frac{\pi}{4}\int_{-\infty}^0 \mathrm{e}^{2t}\cdot t^2\cdot\mathrm{e}^t\mathrm{d}t=\frac{\pi}{4}\int_{-\infty}^0\mathrm{e}^{3t}\cdot t^2\mathrm{d}t$$

$$=\frac{\pi}{4}\int_{-\infty}^0 t^2\mathrm{d}\left(\frac{1}{3}\mathrm{e}^{3t}\right)=\frac{\pi}{12}\mathrm{e}^{3t}\cdot t^2\Big|_{-\infty}^0-\frac{\pi}{6}\int_{-\infty}^0 t\mathrm{e}^{3t}\mathrm{d}t$$

$$=-\frac{\pi}{6}\int_{-\infty}^0 t\mathrm{d}\left(\frac{1}{3}\mathrm{e}^{3t}\right)=-\frac{\pi}{18}\mathrm{e}^{3t}\cdot t\Big|_{-\infty}^0+\frac{\pi}{18}\int_{-\infty}^0\mathrm{e}^{3t}\mathrm{d}t=\frac{\pi}{54}.$$

15.【解析】

$$\frac{\partial u}{\partial x}=\frac{\partial v}{\partial x}\mathrm{e}^{ax+by}+av\mathrm{e}^{ax+by}=\mathrm{e}^{ax+by}\left(\frac{\partial v}{\partial x}+av\right),$$

$$\frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 v}{\partial x^2}\mathrm{e}^{ax+by}+2a\frac{\partial v}{\partial x}\mathrm{e}^{ax+by}+a^2v\mathrm{e}^{ax+by}$$

$$=\mathrm{e}^{ax+by}\left(\frac{\partial^2 v}{\partial x^2}+2a\frac{\partial v}{\partial x}+a^2v\right).$$

同理，

$$\frac{\partial u}{\partial y}=\mathrm{e}^{ax+by}\left(\frac{\partial v}{\partial y}+bv\right),$$

$$\frac{\partial^2 u}{\partial y^2}=\mathrm{e}^{ax+by}\left(\frac{\partial^2 v}{\partial y^2}+2b\frac{\partial v}{\partial y}+b^2v\right).$$

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将上述各式代入原等式，整理得

$$2\frac{\partial^2 v}{\partial x^2}-2\frac{\partial^2 v}{\partial y^2}+(4a+3)\frac{\partial v}{\partial x}+(3-4b)\frac{\partial v}{\partial y}+(2a^2-2b^2+3a+3b)v=0.$$

由题意，令 $4a+3=0$，$3-4b=0$，解得 $a=-\frac{3}{4}$，$b=\frac{3}{4}$．此时，原等式化为

$$\frac{\partial^2 v}{\partial x^2}-\frac{\partial^2 v}{\partial y^2}=0.$$

16.【解析】函数复合关系如图所示，所以

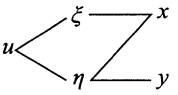
$$\frac{\partial u}{\partial x}=\frac{\partial u}{\partial \xi}\frac{\partial \xi}{\partial x}+\frac{\partial u}{\partial \eta}\frac{\partial \eta}{\partial x}=\frac{\partial u}{\partial \xi}-\frac{\partial u}{\partial \eta},\ \frac{\partial u}{\partial y}=\frac{\partial u}{\partial \eta},$$

$$\frac{\partial^2 u}{\partial x^2}=\frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial \xi}-\frac{\partial u}{\partial \eta}\right)\frac{\partial \xi}{\partial x}+\frac{\partial}{\partial \eta}\left(\frac{\partial u}{\partial \xi}-\frac{\partial u}{\partial \eta}\right)\frac{\partial \eta}{\partial x}$$

$$=\frac{\partial^2 u}{\partial \xi^2}-\frac{\partial^2 u}{\partial \eta\partial \xi}-\frac{\partial^2 u}{\partial \xi\partial \eta}+\frac{\partial^2 u}{\partial \eta^2}=\frac{\partial^2 u}{\partial \xi^2}-2\frac{\partial^2 u}{\partial \xi\partial \eta}+\frac{\partial^2 u}{\partial \eta^2},$$

$$\frac{\partial^2 u}{\partial x\partial y}=\frac{\partial}{\partial \xi}\left(\frac{\partial u}{\partial \xi}-\frac{\partial u}{\partial \eta}\right)\frac{\partial \xi}{\partial y}+\frac{\partial}{\partial \eta}\left(\frac{\partial u}{\partial \xi}-\frac{\partial u}{\partial \eta}\right)\frac{\partial \eta}{\partial y}=\frac{\partial^2 u}{\partial \xi\partial \eta}-\frac{\partial^2 u}{\partial \eta^2},$$

$$\frac{\partial^2 u}{\partial y^2}=\frac{\partial^2 u}{\partial \eta^2}.$$



故 $\frac{\partial^2 u}{\partial x^2}+2\frac{\partial^2 u}{\partial x\partial y}+\frac{\partial^2 u}{\partial y^2}=\frac{\partial^2 u}{\partial \xi^2}=0$，即 $\frac{\partial^2 u}{\partial \xi^2}=0$．

17.【解析】方程 $F\left(x+\frac{z}{y},y+\frac{z}{x}\right)=0$ 两边分别对 x 求偏导，得

$$F_1'\cdot\left(1+\frac{1}{y}\cdot\frac{\partial z}{\partial x}\right)+F_2'\cdot\left(-\frac{z}{x^2}+\frac{1}{x}\cdot\frac{\partial z}{\partial x}\right)=0,$$

则

$$\frac{\partial z}{\partial x}=\frac{y(zF_2'-x^2F_1')}{x(xF_1'+yF_2')}.$$

方程 $F\left(x+\frac{z}{y},y+\frac{z}{x}\right)=0$ 两边分别对 y 求偏导，得

$$F_1'\cdot\left(-\frac{z}{y^2}+\frac{1}{y}\cdot\frac{\partial z}{\partial y}\right)+F_2'\cdot\left(1+\frac{1}{x}\cdot\frac{\partial z}{\partial y}\right)=0,$$

则

$$\frac{\partial z}{\partial y}=\frac{x(zF_1'-y^2F_2')}{y(xF_1'+yF_2')}.$$

所以

$$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=\frac{z(xF_1'+yF_2')-(xF_1'+yF_2')xy}{xF_1'+yF_2'}=z-xy.$$

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18.【答案】(D)

【解析】设在 D 内存在 (x_0,y_0)，使得 $f(x_0,y_0)$ 为正的最大值，则 $f(x_0,y_0)>0$，且

$$f'_x(x_0,y_0)=0,\ f'_y(x_0,y_0)=0.$$

由题设条件得 $f'_x(x_0,y_0)+f'_y(x_0,y_0)=f(x_0,y_0)=0$，矛盾，故选项(A)不正确.

同理，选项(B)也不正确.

故对于任意 $(x,y)\in D$，$f(x,y)\equiv 0$，故 $f(x,y)$ 在 D 的边界上和 D 的内部处处为最大值，处处为最小值.

19.【答案】(B)

【解析】设 $A=\dfrac{\partial^2 f}{\partial x^2}$，$B=\dfrac{\partial^2 f}{\partial x\partial y}$，$C=\dfrac{\partial^2 f}{\partial y^2}$，则 $B>0$，$A+C=0$，于是 $AC\leqslant 0$，故 $AC-B^2<0$，因此在有界闭区域 D 内部无极值点，从而也无最值点. 又由二阶连续偏导数存在可知，$f(x,y)$ 的最小值与最大值一定存在，故必在 D 的边界上取得，选项(B)正确.

20.【答案】(B)

【解析】由题可得　　$f''_{xx}(x,y)=12x^2-2$，$A=f''_{xx}(1,1)=10$，

$$f''_{yy}(x,y)=12y^2-2,\ C=f''_{yy}(1,1)=10,$$

$$f''_{xy}(x,y)=-2,\ B=f''_{xy}(1,1)=-2.$$

于是 $AC-B^2>0$，且 $A>0$，故 $f(1,1)$ 为极小值. 同理可得 $f(-1,-1)$ 也为极小值.

21.【解析】令 $\begin{cases}f'_x(x,y)=3x^2-3y=0,\\ f'_y(x,y)=-3x-2y-1=0,\end{cases}$ 得驻点 $(-1,1)$，$\left(-\dfrac{1}{2},\dfrac{1}{4}\right)$，且

$$f''_{xx}(x,y)=6x,\ f''_{xy}(x,y)=-3,\ f''_{yy}(x,y)=-2.$$

由　　$A=f''_{xx}(-1,1)=-6<0$，$B=f''_{xy}(-1,1)=-3$，$C=f''_{yy}(-1,1)=-2$，

得 $AC-B^2=3>0$，所以 $(-1,1)$ 为极大值点，极大值为 $f(-1,1)=-9$.

由　　$A=f''_{xx}\left(-\dfrac{1}{2},\dfrac{1}{4}\right)=-3$，$B=f''_{xy}\left(-\dfrac{1}{2},\dfrac{1}{4}\right)=-3$，$C=f''_{yy}\left(-\dfrac{1}{2},\dfrac{1}{4}\right)=-2$，

得 $AC-B^2=-3<0$，所以 $\left(-\dfrac{1}{2},\dfrac{1}{4}\right)$ 非极值点.

22.【解析】令 $\begin{cases}f'_x(x,y)=8x^3-8xy-x^6=0,\\ f'_y(x,y)=-4x^2+2y=0,\end{cases}$ 得驻点 $(0,0),(-2,8)$，且

$$f''_{xx}(x,y)=24x^2-8y-6x^5,\ f''_{xy}(x,y)=-8x,\ f''_{yy}(x,y)=2.$$

由 $A=f''_{xx}(0,0)=0,B=f''_{xy}(0,0)=0,C=f''_{yy}(0,0)=2$，得 $AC-B^2=0$. 又在点 $(0,0)$ 附近，

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$$f(x,x^2)=-\frac{1}{7}x^7-x^4=-x^4\left(\frac{1}{7}x^3+1\right)<0,\ f(0,y)=y^2>0(y\neq 0),$$

且$f(0,0)=0$，故$(0,0)$不是极值点.

由$A=f''_{xx}(-2,8)=224,B=f''_{xy}(-2,8)=16,C=f''_{yy}(-2,8)=2$，得$AC-B^2>0$，又$A=224>0$，因此$(-2,8)$为极小值点，极小值为$f(-2,8)=-\frac{96}{7}$，无极大值.

23.【解析】(1) $f'_x(x,y)=y(1+x)e^{x-y}=ye^{-y}(1+x)e^x$，将$y$看作常数，两端同时对$x$积分，有

$$f(x,y)=ye^{-y}\int(1+x)e^x dx=ye^{-y}[xe^x+C(y)].$$

由$f(1,y)=ye^{-y}[e+C(y)]=ye^{1-y}$，得$C(y)=0$，故$f(x,y)=xye^{x-y}$.

(2) 令$\begin{cases}f'_x(x,y)=y(1+x)e^{x-y}=0,\\ f'_y(x,y)=x(1-y)e^{x-y}=0,\end{cases}$解得$\begin{cases}x=0,\\ y=0\end{cases}$或$\begin{cases}x=-1,\\ y=1.\end{cases}$且

$$f''_{xx}(x,y)=y(2+x)e^{x-y},f''_{xy}(x,y)=(1+x-y-xy)e^{x-y},f''_{yy}(x,y)=x(y-2)e^{x-y}.$$

由$A_1=f''_{xx}(0,0)=0$，$B_1=f''_{xy}(0,0)=1$，$C_1=f''_{yy}(0,0)=0$，得$A_1C_1-B_1^2<0$，故点$(0,0)$不是极值点.

由$A_2=f''_{xx}(-1,1)=e^{-2}>0$，$B_2=f''_{xy}(-1,1)=0$，$C_2=f''_{yy}(-1,1)=e^{-2}$，得$A_2C_2-B_2^2>0$，故点$(-1,1)$是极小值点，极小值为$f(-1,1)=-e^{-2}$.

24.【解析】题目相当于求点(x,y)到原点距离的平方$d^2=x^2+y^2$的最值.

设$F(x,y,\lambda)=x^2+y^2+\lambda(x^2+xy+y^2+2x-2y-12)$，令

$$\begin{cases}F'_x=2x+\lambda(2x+y+2)=0,\\ F'_y=2y+\lambda(x+2y-2)=0,\\ F'_\lambda=x^2+xy+y^2+2x-2y-12=0,\end{cases}$$

解得

$$\begin{cases}x=-6,\\ y=6,\end{cases}\begin{cases}x=2,\\ y=-2,\end{cases}$$

$$\begin{cases}x=\frac{3+\sqrt{21}}{3},\\ y=\frac{-3+\sqrt{21}}{3},\end{cases}\begin{cases}x=\frac{3-\sqrt{21}}{3},\\ y=\frac{-3-\sqrt{21}}{3}.\end{cases}$$

从而距离$d=\sqrt{x^2+y^2}$的最大、最小值一定落在以上点当中.

逐一计算可得$d_{\max}=6\sqrt{2}$，$d_{\min}=\frac{2\sqrt{15}}{3}$.

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